Fiscal Year:	FY 2007	**Task Last Updated:**	FY 06/18/2007
PI Name:	Fitts, Robert H Ph.D.		
Project Title:	Effect of Prolonged Space Flight on Human Skeletal Muscle		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Physiology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	53201	**Congressional District:**	4
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	96-HEDS-04
Start Date:	10/01/1998	**End Date:**	09/30/2010
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
Contact Monitor:	McCollum, Suzanne	**Contact Phone:**	281 483-7307
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Flight Program:	ISS		
Flight Assignment:	ISS Increments 5, 6, 7, 9, 10, and 11 (per PI 6/07) Data collection complete, analysis in progress (6/2007) NOTE: Received extension to 9/30/2010 per PI (4/2010) NOTE: Received NCE to 03/30/2010 per J. Dardano/JSC (9/2009) NOTE: Received NCE to 9/30/2009 per PI (1/2009) NOTE: Received NCE to 09/30/2008, per PI (12/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Trappe, Scott (Ball State University) Riley, Danny (Medical College of Wisconsin)		
Grant/Contract No.:	NCC9-116		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Our long term goals are to understand the effects of zero g space flight on the neuromuscular system, and work to develop effective exercise countermeasures. The specific goals of this project are to establish a time course for the structural and functional alterations in limb skeletal muscle with prolonged weightlessness, and from these data calculate the new microgravity steady state that would likely be reached in a trip to Mars. Since we have already made a detailed characterization of how human limb muscles react to short duration (17 days) flight, we propose to study the effects of 3 flights of progressively longer duration. The ideal design would be to study 30, 90, 180 or 30, 60, 90 days. The ideal number of subjects for each flight would be 3 to 4, but we realize from experience that the number available for a given duration flight maybe 1 or 2. This project is designed as a pre- and post-flight study. Calf muscle function will be evaluated using a specially designed torque velocity dynamometer, and biopsies will be taken from the soleus and gastrocnemius muscles at L-45 and immediately post flight (R+0). The effect of prolonged space flight on the functional and structural properties of individual fast and slow fibers will be determined with special emphasis on fiber type specific changes in cell size, peak force and power, the maximal shortening velocity, and the cells resistance to damage. Immunohistochemical studies will be carried out to assess the extent of slow to fast isozyme shifts in myosin, and EM analyses will determine the structural relationship of the thick and thin filament, the degree of myofilament loss, and the integrity of the dystrophin-dystroglycan complex. Microbiochemical studies will establish the extent to which prolonged zero-g affects the substrate and enzyme profile of each fiber type. In conclusion, we will establish the time course for functional and structural change in limb skeletal muscle with prolonged space flight, and establish mechanistic explanations for many of the observed changes. At a minimum we will establish the relative importance of changes in the myosin and actin filament structure vs qualitative changes in the myosin protein, and the relationship of the integrity of the dystrophin-dystroglycan complex to the ability of the cell to resist stretch-induced damage. See also http://www.nasa.gov/
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The primary benefit will be to allow countermeasures to be developed that will prevent the microgravity-induced loss of muscle mass and function, such that, prolonged missions to Mars and beyond can be undertaken. This project will establish the time course of the loss of muscle mass and function with microgravity durations from 30 to 180 days. The results of these studies will have direct benefits to conditions on earth that produce long periods of inactivity, such as, illnesses requiring bed rest, and falls resulting in limb immobilization- the latter is a particularly prevalent problem in the elderly. Of particular importance will be the studies designed to establish the cause of muscle tearing as damage of this sort is a common occurrence in the elderly. Additionally, the results will provide the cellular and molecular data needed to develop effective exercises for slowing the loss of muscle mass that occur with aging.
Task Progress:	This project is an interdisciplinary research effort designed to determine the structural and functional alterations in human limb skeletal muscle during prolonged space flight, and assess the effectiveness of the in-flight exercise countermeasures. Biopsies of the two calf muscles were obtained from 5 astronauts and 5 cosmonauts ~45 days preflight and on landing day after 6 mo aboard the International Space Station (ISS). The study involved crewmembers from increments 5, 6, 7, 9, 10, and 11. Crewmembers self-distributed into those performing high (>200 min/wk) treadmill exercise (HT) and low treadmill exercise (<100 min/wk) (LT). Sample collection and physiological analyses were completed last year. For the past 12 months, we conducted structural studies of muscle size, and intracellular proteins (particularly contractile proteins), lipids and carbohydrates (i.e. glycogen), and biochemical studies of proteins important to the metabolism of muscle. Statistical analyses of the cell physiology data were also performed. The intersubject variation in the percentage of fiber cross-sectional atrophy was striking (3-64% in fibers from the calf muscle). Postflight in both muscles 3 to 48% of slow fibers transitioned into slow/fast (i.e. hybrid fibers) and fast fibers. High volume treadmill exercise was partially protective of the slow fiber. The % decline for the HT group for fiber diameter, force, and power was 9, 23, and 34 %, compared to 26, 43, and 52 % for these variables in the LT group. Following spaceflight central myonuclei indicative of fiber degeneration/regeneration were not significantly different from preflight. The muscle atrophy was associated with a selective loss of contractile filaments, however, the ratios of the two contractile filaments (actin and myosin) were normal postflight. Increased glycogen occurred in all muscles post-flight, but this was likely due to supercompensation of cell glycogen following the last in-flight exercise rather then to a microgravity-induced process. Lipid droplets remained high only in the HT group, indicating availability of lipids may be associated with preservation of fiber size. Metabolic enzymes were relatively unaltered and the oxidative enzymes (reflective of aerobic capacity) were better protected by higher exercise. High volume treadmill exercise appears protective because HT had less atrophy than LT. The severity of atrophy in the LT crewmembers raises concern about their ability to endure long-term spaceflight. Future studies need to address this issue of excess loss of muscle mass and function by developing better inflight exercise programs (particularly high resistance exercise protocols) and improved diets (particularly timing of meals relative to exercise).
Bibliography Type:	Description: (Last Updated: 01/28/2011)
Abstracts for Journals and Proceedings	Fitts RH, Romatowski JG, Peters JR, Trappe S, Gallagher P, Creer A, Costill D, Riley DA. "Human skeletal muscle responses to prolonged spaceflight: Functional capacity of single slow and fast fibers." Presented at Experimental Biology Annual Meeting, Washington, D.C., April 28-May 2, 2007. FASEB Journal 2007;21(6): A952. abstract 771.15. , Apr-2007
Abstracts for Journals and Proceedings	Fitts RH, Colloton PA, Lim WM, Riley DA, Bain JL, Trappe S, Costill D. "Human skeletal muscle responses to prolonged spaceflight: enzyme and substrate adaptations. " Presented at the Experimental Biology Annual Meeting in Washington, D.C., April 28-May 2, 2007. FASEB Journal 2007;21(6):A952. Abstract 771.16. , Apr-2007
Abstracts for Journals and Proceedings	Riley DA, Bain JL Fitts RH, Romatowski JG, Costill D, Trappe S. "Human skeletal muscle responses to prolonged spaceflight: morphological and fiber type changes. " Presented at the Experimental Biology Annual Meeting, Washington, D.C., April 28-May 2, 2007. FASEB Journal 2007;21(6):A953. Abstract 771.18. , Apr-2007

Abstracts for Journals and Proceedings	Trappe S, Peters JR, Gallagher P, Evans H, LeBlanc A, Sinitsyn V, Riley DA, Costil D, Fitts RH. "Human skeletal muscle responses to prolonged spaceflight: whole muscle size and strength. " Presented at the Experimental Biology Annual Meeting, Washington, D.C., April 28-May 2, 2007. FASEB Journal 2007;21(6): A953. Abstract 771.17. , Apr-2007